

SpaceCode 2026

Team “**DOMINATORS**”

Solutions to Theoretical Questions



SCHWARZSCHILD LIVE ELEMENT

$$\hookrightarrow ds^2 = -\left(1 - \frac{2m}{r}\right) dt^2 + \left(1 - \frac{2m}{r}\right)^{-1} dr^2 + r^2 d\phi^2$$

$$ds^2 = d\theta^2 + \sin^2 \theta d\phi^2$$

as $\theta = \frac{\pi}{2}$ (Equatorial Plane)

$$\Rightarrow \sin(\theta) = 1 \quad \& \quad d\theta = 0$$

$$\Rightarrow ds^2 = d\phi^2$$

$$\Rightarrow ds^2 = -\left(1 - \frac{2m}{r}\right) dt^2 + \left(1 - \frac{2m}{r}\right)^{-1} dr^2 + r^2 d\phi^2$$

now $L = \frac{1}{2} g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu$ [given]

$$ds^2 = g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu$$

[By the defn of infinitesimal space time interval]

now we parameterize the worldline with proper time (τ)

$$\Rightarrow x^\mu = x^\mu(\tau)$$

$$dx^\mu = \frac{dx^\mu}{d\tau} \cdot d\tau = \dot{x}^\mu d\tau$$

$$\Rightarrow ds^2 = \left[g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu \right] [d\tau^2] = [2L] [d\tau^2]$$

$$\text{now } dt = \frac{dt}{dz} \cdot dz = \dot{t} dz$$

$$dx = \frac{dx}{dz} \cdot dz = \dot{x} dz$$

$$d\phi = \frac{d\phi}{dz} \cdot dz = \dot{\phi} dz$$

$$\text{and } dt^2 = (\dot{t} dz)(\dot{t} dz) = (\dot{t})^2 dz^2$$

$$dx^2 = (\dot{x})^2 dz^2$$

$$d\phi^2 = (\dot{\phi})^2 dz^2$$

$$\Rightarrow ds^2 = \left\{ -\left[1 - \frac{2M}{r}\right] (\dot{t})^2 + \left[1 - \frac{2M}{r}\right]^{-1} (\dot{x})^2 + r^2 (\dot{\phi})^2 \right\} dz^2$$

$$= 2 f dz^2$$

$$\Rightarrow f = \frac{1}{2} \left[-\left[1 - \frac{2M}{r}\right] (\dot{t})^2 + \left[1 - \frac{2M}{r}\right]^{-1} (\dot{x})^2 + r^2 (\dot{\phi})^2 \right]$$

NOTE:

defⁿ of cyclic :

co-ordinate q is cyclic if $\frac{\partial \mathcal{L}}{\partial q} = 0$

$$\text{as } \mathcal{L} = \frac{1}{2} g_{\mu\nu} \dot{n}^\mu \dot{n}^\nu$$

$$\Rightarrow \frac{d\mathcal{L}}{dt} = 0 \quad \text{and} \quad \frac{d\mathcal{L}}{d\phi} = 0$$

$\Rightarrow t$ and ϕ are cyclic

defn:

Canonical Momentum (p_q) of $q =$

$$p_q = \frac{\partial \mathcal{L}}{\partial \dot{q}}$$

① for t

$$p_t = \frac{\partial \mathcal{L}}{\partial \dot{t}} = \frac{1}{2} \left[-2 \left[1 - \frac{2m}{r} \right] \dot{t} \right]$$

$$\Rightarrow p_t = - \left(1 - \frac{2m}{r} \right) \dot{t}$$

Since t is cyclic $\Rightarrow p_t$ is conserved

$$\mathcal{E} = -p_t = \left(1 - \frac{2m}{r} \right) \dot{t}$$

($\mathcal{E} \rightarrow$ conserved energy per unit mass)

② for ϕ

$$p_\phi = \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = \frac{1}{2} [2\dot{\phi}] = \dot{\phi}$$

as ϕ is cyclic $\Rightarrow p_\phi$ is conserved

$$L = p_\phi = \dot{\phi}$$

($L \rightarrow$ conserved Angular momentum per unit mass)

Schwarzschild spacetime has Rotational Symmetry, the central mass is non-rotating and doesn't have any charge

Rotational Symmetry $\Rightarrow$ Motion depends only on the distance from the center not the orientation.

$\Rightarrow$ if the motion happens on a plane which is tilted in some direction, we can rotate our co-ordinate system so that it appears at the equatorial plane in the new system.

Noether Theorem

every continuous symmetry of action of a physical system $\Rightarrow$ conserved quantity

as there is rotational symmetry $\Rightarrow$ the angular momentum is conserved

$$\text{as } L = r^2 \dot{\phi} \text{ and } L = 0$$

$$\Rightarrow \dot{\phi} = 0$$

$r > 0$, outside the centre

as ϕ doesn't change and only r changes
 $\Rightarrow$ the motion is purely radial
 $\Rightarrow$ the motion takes place on a fixed plane

CONCLUSION:

the motion doesn't depend on orientation and takes place on a fixed plane.

Motion can be taken in the equatorial plane ($\theta = \pi/2$) without the loss of generality.

(B) For timeline geodesics, proper time τ is defined and it satisfies $ds^2 = -d\tau^2$

Normalization conditions:

$$g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu = -1 = 2L$$

as
$$\frac{1}{2} \left[-\left(1 - \frac{2M}{r}\right) (\dot{t})^2 + \left(1 - \frac{2M}{r}\right)^{-1} (\dot{r})^2 + r^2 (\dot{\phi})^2 \right]$$

(Proved in part (A))

$$\Rightarrow -\left(1 - \frac{2M}{r}\right) (\dot{t})^2 + \left(1 - \frac{2M}{r}\right)^{-1} (\dot{r})^2 + r^2 (\dot{\phi})^2 = -1 \quad \text{--- (i)}$$

$$\text{as } -p_t = E = \left(1 - \frac{2M}{r}\right) \dot{t}$$

$$\Rightarrow \dot{t} = \frac{E}{\left(1 - \frac{2M}{r}\right)} \quad \text{--- (ii)}$$

$$\text{as } p_\phi = L = r^2 \dot{\phi} \Rightarrow \dot{\phi} = \frac{L}{r^2} \quad \text{--- (iii)}$$

substitute (ii) & (iii) in eqn (i)

$$-\left(1 - \frac{2M}{r}\right) \frac{(E)^2}{\left(1 - \frac{2M}{r}\right)^2} + \frac{(\dot{r})^2}{\left(1 - \frac{2M}{r}\right)} + r^2 \left[\frac{L}{r^2} \right]^2 = -1$$

$$\Rightarrow -\frac{(E)^2}{\left(1 - \frac{2M}{r}\right)} + \frac{(\dot{r})^2}{\left(1 - \frac{2M}{r}\right)} + \frac{L^2}{r^2} = -1$$

multiply Both sides with $\left(1 - \frac{2M}{r}\right)$

$$-(\dot{r})^2 + (\ddot{r})^2 + \frac{L^2}{r^2} \left[1 - \frac{2M}{r}\right] = - \left[1 - \frac{2M}{r}\right]$$

$$(\ddot{r})^2 = (\dot{r})^2 - \left(1 - \frac{2M}{r}\right) \left(1 + \frac{L^2}{r^2}\right)$$

$$\Rightarrow (\dot{r})^2 = (\ddot{r})^2 + \left(1 - \frac{2M}{r}\right) \left(1 + \frac{L^2}{r^2}\right)$$

$$\text{Let } v^2(r) = \left(1 - \frac{2M}{r}\right) \left(1 + \frac{L^2}{r^2}\right)$$

$$\Rightarrow (\dot{r})^2 = (\ddot{r})^2 + v^2(r)$$

Hence Proved.

③ given $L=0$ and the spaceship start from rest at $r=R$ as the it start from Rest $\Rightarrow \dot{r}=0$ at $r=R$

$$\dot{r}^2 + v^2(r) = \epsilon^2 \quad \text{--- (i)}$$

$$\text{where } v^2(r) = \left(1 - \frac{2M}{r}\right) \left(1 + \frac{L^2}{r^2}\right)$$

[derived in (b)]

at $r=R$

$$(0)^2 + \left(1 - \frac{2M}{R}\right) \left(1 + \frac{0}{R^2}\right) = \epsilon^2$$

$$\Rightarrow \epsilon^2 = 1 - \frac{2M}{R} \quad \text{--- (ii)}$$

when the distance from the centre is R :

$$(\dot{r})^2 + \left(1 - \frac{2M}{r}\right) = \epsilon^2 \quad \text{--- (iii)}$$

substitute (ii) in (iii)

$$(\dot{r})^2 + \left(1 - \frac{2M}{r}\right) = \left(1 - \frac{2M}{R}\right)$$

$$\Rightarrow (\dot{x})^2 = 2m \left[\frac{1}{x} - \frac{1}{R} \right] \quad \text{--- (14)}$$

$$\dot{x} = \frac{dx}{dz}$$

$$\dot{x} = \pm \sqrt{2m \left[\frac{1}{x} - \frac{1}{R} \right]}$$

we will take the -ve sign because the value of x decreases as z increases.

$$\Rightarrow \dot{x} = - \sqrt{2m \left[\frac{1}{x} - \frac{1}{R} \right]} = \frac{dx}{dz}$$

(Differential Eqn)

Taking Substitution that produces
 $\hookrightarrow x(X) = \frac{R}{2} (1 + \cos X) = R \cos^2 \left(\frac{X}{2} \right)$ (Cycloid)

$$\left[\cos X = 2 \cos^2 \left(\frac{X}{2} \right) - 1 \right]$$

$$\frac{dx}{dz} = - \sqrt{\frac{2m}{R} \left[\frac{1}{\cos^2 \left(\frac{X}{2} \right)} - 1 \right]} = \sqrt{\frac{2m}{R} \tan^2 \left(\frac{X}{2} \right)}$$

$$\frac{dx}{dz} = - \sqrt{\frac{2m}{R}} \left(\tan \frac{X}{2} \right)$$

$$\text{as } r(x) = \frac{R}{2} [1 + \cos x]$$

$$dr = -\frac{R}{2} \sin x \, dx$$

$$dZ = \frac{dr}{-\sqrt{\frac{2M}{R}} \tan\left(\frac{x}{2}\right)}$$

$$dZ = \frac{+ \frac{R}{2} \sin(x) \, dx}{+ \sqrt{\frac{2M}{R}} \tan\left(\frac{x}{2}\right)}$$

$$\left[\sin(x) = 2 \sin\left(\frac{x}{2}\right) \cos\left(\frac{x}{2}\right) \right]$$

$$dZ = \frac{R\sqrt{R}}{2\sqrt{2M}} \cdot 2 \cos^2\left(\frac{x}{2}\right) dx$$

$$dZ = \sqrt{\frac{R^3}{8M}} (\cos(x) + 1) dx$$

$$\text{as } r(x) = \frac{R}{2} [1 + \cos x]$$

$$\Rightarrow \text{at } \begin{array}{l} x=0 \\ x=\pi \end{array} \quad \left| \quad \begin{array}{l} r=R \text{ (start of motion)} \\ r=0 \text{ (end of motion)} \end{array} \right.$$

consider $Z=0$ at the start of motion

consider that the proper time is τ at some X

$$d\tau = \sqrt{\frac{R^3}{8m}} [1 + \cos x] dx$$

Integrate Both Sides

$$\int_0^\tau d\tau = \int_0^x \sqrt{\frac{R^3}{8m}} [1 + \cos x] dx$$

$$\Rightarrow \tau = \sqrt{\frac{R^3}{8m}} [x + \sin(x)] \Big|_0^x$$

$$\tau = \sqrt{\frac{R^3}{8m}} [x + \sin(x)]$$

↳ is the Parametric soln of the Differential Eqn

time taken to fall from $r=R$ to $r=0$
[$x=0$ to $x=\pi$] is

$$\tau = \sqrt{\frac{R^3}{8m}} [\pi + \cancel{\sin(\pi)}]$$

$\downarrow$
0

$$\tau = \pi \sqrt{\frac{R^3}{8m}} \text{ sec}$$

(b)

$$\frac{d\sigma}{dt} = \frac{d\sigma}{d\tau} \cdot \frac{d\tau}{dt} = \frac{\dot{\sigma}}{\dot{\tau}}$$

$$\text{as } \varepsilon = \left(1 - \frac{2m}{\sigma}\right) \dot{\tau} \Rightarrow \dot{\tau} = \frac{\varepsilon}{\left(1 - \frac{2m}{\sigma}\right)}$$

$$\frac{d\sigma}{dt} = \frac{\dot{\sigma}}{\dot{\tau}} = \frac{\dot{\sigma} \left(1 - \frac{2m}{\sigma}\right)}{\varepsilon}$$

$$\tau_* = \sigma + 2m \ln \left| \frac{\sigma}{2m} - 1 \right|$$

$$\text{and } \frac{d\sigma}{dt} = \frac{\left(1 - \frac{2m}{\sigma}\right) \dot{\sigma}}{\varepsilon}$$

$$\frac{d\sigma_*}{dt} = \frac{d\sigma}{dt} + \cancel{2m} \cdot \frac{1}{\left(\frac{\sigma}{2m} - 1\right)} \cdot \frac{1}{\cancel{2m}} \frac{d\sigma}{dt}$$

$$\frac{d\sigma_*}{dt} = \frac{d\sigma}{dt} \left[1 + \frac{1}{\left(\frac{\sigma}{2m} - 1\right)} \right]$$

$$= \frac{\left(1 - \frac{2m}{\sigma}\right)}{\varepsilon} \left[1 + \frac{1}{\left(\frac{\sigma}{2m} - 1\right)} \right] \dot{\sigma}$$

$$= \frac{\dot{\sigma}}{\varepsilon} \left(\frac{\cancel{\sigma} - \cancel{2m}}{\cancel{\sigma}} \right) \left(\frac{\cancel{\sigma}}{\cancel{2m}} \right) \left(\frac{\cancel{2m}}{\cancel{\sigma} - \cancel{2m}} \right)$$

$$\Rightarrow \frac{d\sigma_*}{dt} = \frac{\dot{\sigma}}{\varepsilon}$$

$$(\ddot{r})^2 = \epsilon^2 - v^2(r)$$

$$\text{where } v^2(r) = \left(1 - \frac{2M}{r}\right) \left(1 + \frac{L^2}{r^2}\right)$$

(given) as $r \rightarrow 2M^+$

$$1 - \frac{2M}{r} \rightarrow 0^+ \text{ and as } 1 + \frac{L^2}{r^2} \text{ is finite}$$

$$\Rightarrow v^2(r) \rightarrow 0^+$$

$$\Rightarrow \epsilon^2 - v^2(r) \approx \epsilon^2$$

$$\Rightarrow (\ddot{r})^2 \approx \epsilon^2$$

$$\Rightarrow \ddot{r} \approx \pm \epsilon$$

as the object is falling towards singularity, the value of r decreases with t

$$\Rightarrow \ddot{r} < 0$$

$$\Rightarrow \boxed{\ddot{r} \approx -\epsilon}$$

$$\text{as } \frac{d\ddot{r}}{dt} = \frac{\ddot{r}}{\epsilon} \approx -1$$

$$\frac{d\ddot{r}}{dt} \approx -1 \Rightarrow \ddot{r} \approx -dt$$

Integrate Both sides

$$\ddot{r} \approx -t + A, \text{ where } A \text{ is a constant}$$

let $r_* = 2M + \epsilon$ where $\epsilon > 0$ and $\epsilon \ll 2M$

$$r_* = r + 2M \ln \left| \frac{r}{2M} - 1 \right|$$

$$\frac{r}{2M} - 1 = \frac{\epsilon}{2M} > 0$$

$$r_* = r + 2M \ln \left(\frac{\epsilon}{2M} \right)$$

$$r + 2M \ln \left(\frac{\epsilon}{2M} \right) = -t + A$$

$$2M \ln \left(\frac{\epsilon}{2M} \right) = -t - \underbrace{(2M + \epsilon)}_{\text{constant}} + A$$

$$\ln \left(\frac{\epsilon}{2M} \right) = \frac{-t}{2M} + \underbrace{\frac{C'}{2M}}_{\text{constant}}$$

$$\ln \left(\frac{\epsilon}{2M} \right) = \frac{-t}{2M} + D$$

take exponentiation of Both sides

$$\frac{\epsilon}{2M} = B e^{\frac{-t}{2M}}$$

$$\epsilon = \underbrace{2MB}_{\text{constant}} e^{\frac{-t}{2M}}$$

$$\epsilon = C e^{-t/2M} \quad \text{as} \quad \epsilon = r - 2M$$

$$\Rightarrow r(t) \approx 2M + C e^{-t/2M}$$

the t used is Schwarzschild
co-ordinate time.

it is the amount of time observed
by an observer at ∞ who observes
the object falling towards the
Event Horizon

Schwarzschild radius = $\frac{2GM}{c^2}$

$$r(t) \approx \frac{2GM}{c^2} + C e^{-\frac{c^3 t}{2GM}}$$

$$r(t) \approx \frac{2GM}{c^2} + C \exp\left(-\frac{c^3 t}{2GM}\right)$$

③

asymptotic $\propto e^M$ (from ②)

$$r(t) \approx 2M + C e^{-t/2M}$$

where t is Schwarzschild co-ordinate time

$$\text{as } t \rightarrow \infty \Rightarrow r(t) \rightarrow 2M$$

$\Rightarrow$ Achilles won't reach the tortoise (at $r = 2M$) in finite amount of time (t)

Schwarzschild co-ordinate time (t)

$\hookrightarrow$ is the time measured by an observer at ∞ who observes the Achilles falling towards event horizon

Proper time (τ)

$\hookrightarrow$ it is the amount of time measured by Achilles during his fall towards the event horizon.

[in part ②]

we found that

$$\tau \text{ is } \sqrt{\frac{R^3}{8M}} \pi$$

while travelling from R to 0. Achilles would pass the Event Horizon also during this fall as $\tau_{R \rightarrow 0}$ is finite $\Rightarrow$ time taken to

reach the Event Horizon is also finite.

Thus:

the proper time (τ) is FINITE
where as the
Schwarzschild coordinate time (t)
is INFINITE

CONCLUSION:

Achilles reach the Tortoise (in physical sense) in finite amount of time, but the time stretches for the observer at ∞ because of which τ is finite and t is infinite

Why 't' stretches?

$$\text{as } \frac{dt}{d\tau} = \frac{1}{1 - \frac{2M}{r}}$$

when $r \rightarrow 2M^+$ $t \rightarrow \infty$

$\therefore$ a small amount of Achilles's proper time (τ) correspond to a very huge amount of Schwarzschild coordinate time (t)